Physio Questions

1. The characteristic of loop of Henle include

a. Descending limp is not permeable to water

b. Urine because more dilute as it reaches the tip of the loop

c. Ascending limb is permeable to water

d. Thin ascending limb urea is actively transported to interstitial fluid

e. Na is reabsorbed by Na/k/2cl in ascending limb

Ans. E

2. The following is true about vasa recta ?

a. Descending vasa recta reabsorbs water from interstitial fluid

b. The descending capillaries loose Na and urea

c. Plasma becomes hyperosmotic as it descend in Vasa recta

d. Ascending capillaries loose water to interstitial fluid

e. They are reasonable for countercurrent multiplayer mechanism

Ans. C

3. The function of renal PCT include

a. Reabsorption of all bicarbonate filtered at glomeruli

b. Production of renin

c. Reabsorption of 80% of filtered Hco3

d. Reabsorption of bicarbonate resulting in fixed acid secretion

e. hydrogen ion secreted by facilitated diffusion.

Ans. C

4. For those substance that are actively reabsorbed or secreted , the highest limit of the amount that can be transported per unite time by the kidney

a. Depends on the rate of at which the transport mechanism operate

b. Is directly related to plasma conc of the substance

c. Depends on the tubler transport maximum

d. It depends upon the tubler load

e. Depends of the GFR

Ans: c

5. If the clearance of substance X is greater than the clearance of inulin it is most likely that the substance X is

a. Bound to tubular protein

b. Bound to plasma protein

c. Secreted

d. Reabsorbed

e. Neither secreted or reabsorbed

Ans: c

6. Renin is produced and secreted from the granules located in the

a. Macula densa

b. Mesangial cells

c. Intercalated cells

d. Afferent arterioles

e. Interstitial cells

Ans: d

7. The tonicity of urine as it enters the renal collecting duct is

a. Isotonic

b. Hypotonic or isotonic but never hypertonic

c. Hypotonic

d. Hypertonic

e. Hypertonic or isotonic but never hypotonic

Ans: c

8. In a healthy adult man the GFR is expected to be

a. 225 ml/min

b. 325 ml/min

c. 125 ml/min

d. 25 ml/min

e. 180 ml/min

Ans: c

9. The following factors which increase they will increase the GFR Except

a. Renal blood flow

b. Afferent arterioles diameter

c. Efferent arterioles diameter

d. Permeability of glomerular membrane

e. Capillary surface area

Ans: c

10. The substance that used to measure renal plasma flow should be

a. Bound to plasma protein

b. Easily reabsorbed

c. Highly secreted

d. Metabolized by the body

e. Stored only in the kidney

Ans: c

11. Renal blood flow is

a. increased when renal sympathetic nerves are stimulated

b. Autoregulated when mean arterial blood pressure is between 60-180

c. Measured by urea clearance.

d. about 2 liter in a normal man

e. Decreased when the blood flow is at a site of macula densa decreased.

Ans: b

12. GFR would be increased by

a. A decrease of afferent arteriole measure

b. A decrease in renal blood flow

c. A decrease in efferent arteriole resistance

d. Decrease in the conc of plasma protein

e. Obstruction of ureters

Ans: d

13. Potassium secretion in DCT will be increased if there is increase in

a. Plasma hydrogen

b. Plasma volume

c. Plasma sodium conc

d. Plasma potassium conc

e. Plasma calcium conc

Ans: d

14. The following is true about glucose reabsorption in the renal tubules

a. Transport maximum is 80 mg/min

b. The transport maximum is dependent on the amount of glucose filtered

c. It's mainly by symport with k+

d. It's reduced with metabolic inhibitors, that reduce the activity of Na/K Atpase pump.

e. About 50% of filtered glucose is reabsorbed

Ans: d

15. Water reabsorption in the collecting duct

a. Is dependent on aldosterone

b. Is increased by ADH

c. Has the largest percentage of total water reabsorption

d. Is facilitated by Na reabsorption

e. Independent of osmotic difference across tubules

Ans: b

16. The GFR can be calculated by all of the following Except

a. Ph clearance b. Creatinine clearance

c. Urea clearance d. Inulin clearance

Ans: a

17. Stimulator for increasing renal renin secretion is an increase in which of the following

a. Mean blood pressure

b. GFR

c. Sympathetic nerve activity

d. Circulating angiotensin 2

e. Arterial neutritic peptide secretion

Ans: c

18. Which of the following is most likely to cause an increase in GFR

a. Contraction of mesangial cells

b. Blockage of the ureters

c. Release of renin from Juxtaglomerular apparatus

d. Dilatation of afferent arterioles

Ans: d

19. Normal GFR

a. 25 ml/min b. 50 ml/min

c. 75 ml/min d. 100 ml/min

e. 125 ml/min

Ans: E

20. A 32-year-old man complaining of fatigue and muscle weakness is seen by his physician. Blood tests reveal a serum glucose level of 325 mg/dL and serum creatinine of 0.8 mg/dL. Results of a 24-hour urine analysis are as follows:

- **Total volume = 5L**
- **Total glucose = 375 g**
- **Total creatinine = 2.4 g**

The patient's GFR is approximately which of the following?

a. 75 ml/min b. 100 ml/min

c. 125 ml/min d. 200 ml/min

e. 275 ml/min

Ans. D

21. Renal blood flow is kept constant by auroregulation in mean systemic blood pressure is range of

a. 0-90 mmHg

b. 60-180 mmHg

c. 200- 250 mmHg

d. All of the above

Ans: b

22. GFR may be decreased by

a. ADH b. Renal oedema

c. Hypertension d. Sympathetic stimulation

Ans: d +b

23. Sodium reabsorption is coupled with the following substance in PCT

a. Hydrogen ion b. Amino acid

c. Glucose d. All of the above

Ans: D

24. The GFR will increase if

a. Circulating blood volume increase

b. The afferent arteriole resistance increase

c. Efferent arteriole resistance decrease

d. The plasma protein conc decrease

Ans: D

25. The greatest amount of hydrogen ion secreted by the proximal tubule is associated with :

a. excretion of K ion b. excretion of H ion

c. reabsorption if Ca ion d. reabsorption of Hco3 ion

Ans: d

26. The ability of the kidney to excrete a concentrated urine will increase if:

a.the permeability of proximal tubule to water decreased

b. The rate of blood flow through the medulla decreases

c. the rate of flow through loop of Henle increases

d. Activity of Na/K pump in loop of Henle increase

Ans: B

27. Glomerular filtration rate would be increased by:

a. Constriction of the afferent arteriole

b. A decrease in afferent arteriolar pressure

c. Compression of the renal capsule

d. A decrease in the concentration of plasma protein

Ans: d

28. In the distal tubules, Na reabsorption is increased directly by increased:

a. sympathetic nerve stimulation of the kidney

B. atrial natriuretic hormone secretion

c. antidiuretic hormone secretion

d. aldosterone secretion

Ans: d

29. which of the following substances will be more concentrated at the end of proximal tubule than at the beginning of proximal tubule

a- glucose b- creatinine

c- sodium d- bicarbonate

Ans: b

30. the amount of potassium excreted by the kidney will decrease if

a- distal tubular flow increase

b circulating aldosterone level increase

c- dietary intake of potassium increase

d- sodium reabsorption by distal nephron decrease

Ans: d

31. The effect of ADH on the kidney is to :

a) increase the permeability of the distal nephron to water

b) increase the excretion of Na+

c) increase the excretion of water

d) increase the diameter of renal artery

Ans: A

32. When a person is dehydrated, hypotonic fluid will be found in the :

a) glomerular filtrate

b) proximal tubule

c) descending lobe of loop of henle

d) distal convoluted tubule

Ans: D

33. glucose is reabsorbed from apical membrane of proximal convoluted tubule using

a- passive diffusion b- endocytosis

c- facilitated diffusion d- secondary active transport

e- primary active transport

Ans: d

34. hydrogen ion is Not secreted in the following part of the nephron

a- proximal convoluted tubule b- loop of henele

c- distal convoluted tubule d- collecting duct

E- non of the above

Ans: E

35. which of the following injury will lead to large volume, hypotonic bladder ?

a- transection of affront pelvic nerve from bladder

b- transection of both afferent pelvic nerve and efferent parasympathetic nerve to bladder

c- transection of brain stem above midbrain

d- transection of brain stem above pones

Ans: b

36. Regarding water reabsorption:

a. The ascending limb of the loop of Henle is impermeable to water.

b. Maximum urine osmolarity is about 800 mosm/L.

c. The CD is freely permeable to water even in the absence of ADH.

d. The DCT is permeable to water.

Ans: a

37. In renal tubules, bicarbonate is responsible for buffering hydrogen ion in :.

A) proximal convoluted tubules

B) loop of henle

C)Distal convoluted tubules

D) collecting tubules

Ans: A

38. Hydrogen ion secretion in proximal convoluted tubules uses:

A) passive diffusion

B) proton pump

C)H+,K+ exchange

D)H+, Na+ exchange

Ans: D

39. Regarding the juxtaglomerular apparatus:

a. Lacis cells secrete renin.

b. The macula densa is modified afferent arteriole endothelium.

c. Renin is secreted in response to reduced tubular pressure.

d. Juxtaglomerular cells are located in the walls of the afferent arteriole of the glomerulus

Ans: d

40. Which of the following statement is True

A. Thin ascending limb of loop of Henle has Na-k-2cl active transport

B. Thin descending limb of loop of Henle is impermeable to water

C. Osmolarity is greatest at the beginning of loop of Henle

D. Water diffuse out of blood vessels from the ascending limb of vasa recta into it's descending limb

E. None of the above

Ans: E

41. The following substance leads to increased sodium reabsorption in kidney

A. Aldosterone

B. Vasopressin

C. ANP

D. PgE2

E. All of the above

Ans: A

42. The volume of plasma needed each minute to supply a substance at the rate at which it is excreted in the urine is known as the ?

a. Diffusion constant of substance

b. Clearance of substance

c. Execretion ratio of the substance

d. Tubular mass of the substance

Ans: B

43. in a healthy individual breathing spontaneously which of the following pressure is positive with reference to atmospheric pressure?

a. Alveolar pressure during inspiration

b. Alveolar pressure during expiration

c. Intrapleural pressure during inspiration

d. Intrapleural pressure during expiration

Ans: B

44. The maximum amount of gas that can be exhaled after full inspiration is called

a. Expiratory reserve volume

b. Vital capacity

c. Total lung capacity

d. Functional residual capacity

Ans: B

45. The largest volume of gas that can be moved into and out of the lung in 1 minute by voluntary effort called.....

a. Respiratory minute volume

b. Minute ventilation

c. Maximum voluntary ventilation

d. Vital capacity

Ans: C

46. Intrapleural pressure is normaly negative because ...

Ans: chest wall and lung recoil in opposite direction at FRV

47. What is the effect of 2,3 BPG on affinity of Hg for oxygen?

a. Increase affenity

b. decrese affenity

Ans: B

48. which of the following has the highest p50?

a. Myoglobin b. Hbf c. Hba

Ans: C

49. All of the following shift oxyhemoglobin dissosiation curve to the right except

a. Co

b. Co2

c. Low ph

Ans : A

50. decrease in o2 affenity of Hb called

a. Boher effect

b. Handal effect

Ans: A

51. Which of the following reduce synthesis of 2,3 BPG ?

a. Acidosis

b. At exercises

c. Anemia

d. In high attetiude

Ans: A

52. The amount of oxygen dissolved in plasma ??

Ans: Po2

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